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AI Engineering Lead · Development leader, 10 years

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In one line

A development leader with 10 years of experience, turning research- and pilot-stage technology into products people actually use and pay for. I design and write the code myself, and have led teams from a 3-person dev unit to a 50-person research division.

Core strengths

01 From pilot technology to commercial product

- Took [Alan](#), an LLM service piloted by a research team, to a Pro subscription for its first paid revenue, and planned the enterprise plan — laying the groundwork for institutional contract revenue
- Built [Dr.Meta](#) and [Meta.CRO](#) on the output of [MX Studio](#), a 3D web component tool — in commercial operation at cancer centers nationwide

02 Designing AI systems hands-on, proving them in code

- Designed a code-generation pipeline combining rule-based modules with AI models and built its core (PoC) myself → officially launched with the team as [ProtoPie MCP](#) in October 2026
- Designed and built [Decision Graph](#), an LLM pipeline that extracts decisions, on my own — over 80% reliability on a golden set from real projects

03 Deciding by measurement, not gut feel

- Model choices made against labeled test cases — rejected a switch to a lighter model that got only 17 of 24 right, and restructured the calls to cut run cost from \$10.2 to \$7.6
- Set development priorities from traffic flows and per-feature usage in Google Analytics and Amplitude

04 Leadership sized to the team

- Built and led teams from a 3-person dev unit to a 50-person research division; 3 group-wide awards (Maestro · Super Leader · Technology Innovation Award)
- Running an agile team of 4–5 AI engineers and app developers — build small, validate fast

Experience

Studio XID (ProtoPie) · Head of Product Dev Team

2026.01.06 - Present

Product Dev Team

LLM pipeline for decision extraction (Slack·Notion → decision records with evidence)

2026.08 - 2026.10

Design & development (solo) · Internal tool: [Decision Graph](#)

- Designed and built, solo, an LLM pipeline that extracts who decided what and why from Slack and Notion, together with the source text as evidence
- Validated PoC results using one real project from the past year as the golden set — confirmed reliability of 80%+
- Verified and reviewed the feared technical risks firsthand through the PoC (judging by labeled test cases instead of a reliability %, one-way records, preventing duplicate entries on re-runs)
- Chose the two-tier inference structure (first-pass classification: lightweight model; extraction and validation: high-capability model) by comparing options on the same labeled test cases

- Logged time and cost per request to find bottlenecks and improved the call structure — on the same data, 1,289 calls · \$10.2 → 336 calls · \$7.6; re-running only the changed parts: 30 calls · \$0.7
- Checked quality on every change with 23 labeled test cases (with a cap on evaluation cost)

AI-based code generation system (PoC → productization)

2026.01 - 2026.10

Head of Product Dev Team · Product: [ProtoPie MCP](#) (Dev View · Code MCP) official launch

- Led the team for the overall design and development of the AI-based code generation system
- Ran an agile team of 4–5 AI engineers and app developers
- Designed the code generation pipeline combining rule-based modules with AI model stages (led directly)
- Designed training for models that support image-based and layout (structure)-based code conversion, and built training data including labeling (training done by the AI engineers)
- Designed the MLOps pipeline from data cleaning through training and deployment
- Built the cross-platform component code generation pipeline for AOS, iOS and Web
- Developed the core pipeline directly (PoC) → productized it with the team → official launch as ProtoPie MCP in October 2026 (Dev View · Code MCP, code generation for 7 frameworks)

ESTsoft · Product Owner

2025.04.22 - 2026.01.05

AI Agent Lab / Alan Development Part / PO

AI search engine service (LLM-based agentic AI) Advancing 'Alan' (ESTsoft)

2025.04 - 2026.01

Product Owner · Product: Alan

- Led a service in trial operation at the research organization through commercial launch and monetization
- Led the development part for an LLM-based agentic AI app service, working as Product Owner
- Led a part made up of AI researchers and app developers (FE, BE)
- Operated and managed separate Dev, QA, Stage and Release servers
- Architected the Azure, Fast API, React and LangGraph structure
- Ran quality testing and collaborated with other in-house development, quality and infrastructure teams (QA, authentication, payments, infrastructure, etc.)
- Handled feature planning, scheduling and risk management for the LLM-based AI app service
- Independently developed image search, YouTube search & summarization, report generation and slide generation features plus specialized agents (led slide generation from research and planning onward)
- Set up the payment system process and the paid product policy
- Deployed the deep research and slide generation services, launched the Pro subscription, and developed an internal back-office operations tool
- Pro subscription brought the service's first paid revenue; building on it, planned an enterprise product (later the basis for institutional contract revenue)
- In commercial operation ([myalan.ai](#))

TmaxMetaAI · Project Leader, Development Team

2022.07.01 - 2025.04.14

Meta AI Research Division / Researcher (Head of Office)

Software for building 3D web components Development of 'MX studio' and productization based on it 2022.07 - 2025.02

Project Leader · Product: MX studio

- Led the MX studio engine team
- Led a task force team of planners, designers, developers and QA
- Led design/development of events and actions for 3D objects
- Led design/development of the physics engine, real-time rendering and post-processing features
- Led design/development of a node-based visual code system
- Ran an external beta test

Metaverse platform used by cancer centers nationwide Development of 'Dr.Meta' (Korea Smart Healthcare Association) 2024.06 - 2025.01

Project Leader · Product: Dr.Meta

- Led a task force team of planners, designers, developers and QA
- Led design/development of the integration between a React-based web app and a Unity app
- Led design/development of the admin page and permission features
- In commercial operation at cancer centers nationwide (web.drmeta.kr)

3D-based virtual simulation training platform for clinical trials Development of 'Meta.CRO' (Korea Smart Healthcare Association) 2023.06 - 2025.01

Project Leader · Product: Meta.CRO

- Led a task force team of developers and QA
- Designed and released a 3D render engine library (set up a private npm environment)
- In commercial operation at cancer centers nationwide

Online marketplace platform for contact lenses Development of 'NaenunN' (Pixelro) 2024.03 - 2024.09

Project Leader · Product: NaenunN

- Led a task force team of developers and QA
- Developed, together with the partner company, an online contact lens purchase and delivery service first permitted under a regulatory sandbox special exemption for demonstration
- Led design/development of a cross-platform app from a single core codebase using React and React Native
- Led builds and releases to meet Android and iOS review requirements
- In commercial operation (nenoonn.mycafe24.com)

GIS-based Samsung data center information system Development of 'GIS S-DCIS' (Samsung C&T) 2023.12 - 2024.08

Project Leader · Product: GIS S-DCIS

- Communicated and collaborated with the planning team, the business team and another company's research team
- Built a spatial function server and DB using GeoServer and PostGIS
- Researched Openlayers open-source technology to build a map platform on GIS data
- Used public data and handled policy issues
- Troubleshoot issues in migrating source code to a closed network and setting up an on-premise environment

- Co-developed with Yeonwoo Technology development researchers and delivered (B2B)

Real-time multi-sync server Research on 'MVS: Metaverse Server' (internal project) 2023.01 - 2024.09
Project Leader · Product: MVS

- Led the R&D team; researched and developed C++-based server engine technology
- Designed client libraries (Javascript, C#)
- Designed the service flow
- Applied the researched technology as the multiplayer server for Dr.Meta's medical staff conferences

TmaxAI · Project Leader 2020.07.25 - 2022.06.30
Meta Vision Research Division / Researcher (Head of Office, concurrently Division Head)

Metaverse education platform Development of 'Mint' 2021.07 - 2022.06
Project Leader · Project: Mint

- Planned the product using Gather Town as a benchmark
- Used open-source technologies including the Phaser game engine, WebRTC, WebGL, React and MobX
- Delivered to Kongju National University as a metaverse education platform

Advanced technology research Deep Learning, 3DMM, SLAM 2020.07 - 2021.06
Project Leader · Research: Digital human and metaverse space research

- As advanced technology research, studied generating digital humans capable of Deep Learning-based text-to-speech
- As advanced technology research, studied generating metaverse spaces using Deep Learning-based SLAM 3D spatial technology
- Used Blender and WebGL open source
- Demonstrated at the demo booth of the 2020 Tmax Day conference

TmaxA&C · Product Engineer 2017.01.02 - 2020.07.24
Graphics Research Division / Researcher (Team Lead)

Collaboration solution Development of 'WAPL' 2019.01 - 2020.06
Product Engineer · Product: WAPL

- Developed calendar and messenger features (frontend)
- In commercial operation by TmaxWAPL

Linux-based product Development of 'TmaxOS' 2017.06 - 2018.12
Product Engineer · Product: TmaxOS

- Developed in C++
- Fixed bugs in the OS personal settings section (frontend)
- Developed a video player and refactored its code

Integrated system interface solution software Structure analysis of 'Anylink' 2017.02 - 2017.06
Structure analysis & refactoring design · Product: Anylink

- Took part in structure analysis and refactoring design of the integrated system interface solution
- Worked with the engine research team to identify usability and quality improvement points
- In commercial operation by TmaxSoft ([Product overview](#))

Skills

AI · LLM	Python · LLM API · LangGraph · MCP · RAG · Computer Vision
Backend · Infra	Node.js · FastAPI · Java · Spring · C/C++ · Docker · Jenkins · AWS · Azure
Frontend · Cross-platform	TypeScript · JavaScript · React · React Native · Three.js · Unity
Collaboration · Analytics	Git · Jira · Notion · Figma · Google Analytics · Amplitude
Platforms	Windows · Linux · iOS · Android

Education

KAIST, Culture Technology · M.S. in Culture Technology (Master's degree)	2013.09 - 2016.02 (Graduated)
3.5 / 4.3 · Human-computer interaction, computational design, game studies, IoT	
Ajou University (College of Information Technology), Digital Media · B.S. in Digital Media (Bachelor's degree)	2007.03 - 2013.08 (Graduated)
3.76 / 4.5 · Computer programming, computer architecture, object-oriented programming, information design	

Awards & activities

Award

2023.01	Annual Technology Innovation Award — Received the outstanding employee award titled 'Annual Technology Innovation Award' at TmaxMetaAI (Given once a year to about 10 people out of all employees across the group)
2022.06	Super Leader Award — Received the excellence award titled 'Super Leader' at TmaxAI (Given twice a year to about 5 people out of all employees across the group)
2022.01	Maestro Award — Received the top award titled 'Maestro' at TmaxAI (Given once a year to about 3 people out of all employees across the group)

Paper & research

2016.02	Master's thesis Title: A Study on Mobile Game Design Strategies that Enhance Enjoyment and Self-Efficacy to Promote Everyday Physical Activity
2016.01.29	Conference paper 2 Title: A Study on Mobile Game Design Using Gamification and Symbolization Strategies to Promote Physical Activity in Everyday Life Organizer: HCI Society of Korea · Authorship: First author · Pages: Vol.2016 No.1 [2016] p170-177 (Korean)
2016.01.28	Conference paper 1 Title: Gamification Can Affect the Sustainability of The Wearable Device for Physical Activity? : User Case Study Organizer: HCI Society of Korea · Authorship: First author · Pages: Vol.2016 No.1 [2016] p451-455 (English)
2014.11	Journal paper Title: An Exploratory Study of Factors Affecting the Usability and Sustainability of Wearable Devices for Health Management - A Review of User Experience from a Convergence Perspective of Technology, Psychology and Interaction Authorship: Third author · Organizer: Korean Institute of Information Scientists and Engineers, Communications of the KIISE · Pages: Vol. 32, No. 11 (Issue 306), p37-45 (Korean)

Certifications & languages

2025.01.12 **OPic Speaking** — English (IM1) · ACTFL
2010.09.03 **MOS Master 2007** — Microsoft Corporation

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